Gene Expression Analysis of RBM5 and CFLAR

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1 Introduction

Gene expression analysis is an important tool to gain understanding for how genes function and identifying patterns in gene expression. It allows for researchers to effectively study disease, understand cell identity, and track body responses. This particular project examines the genes RBM5 and CFLAR using RNA-sequencing count data and associated sample metadata.

RNA Binding Motif Protein 5, RBM5, is known to be a potential tumor suppressor gene that produces a protein in RNA processing. RBM5 is highly expressed in the brain, endocrine system, and respiratory system. Furthermore, it plays a role in apoptosis (programmed cell death) and cell-cycle arrest (National Center for Biotechnology Information [NCBI], 2026). RBM5 may be involved with the prevention of tumor transformation and progression of cancers such as lung cancer. On the other hand, CASP8 and FADD-like apoptosis regulator, CFLAR, is a gene that also plays a role in regulating apoptosis. CFLAR also aids in controlling inflammation and the activity of immune cells. Within the tumor microenvironment, higher CFLAR expression may help explain why some cancer cells resist treatment (Zhang et al., 2025).

The analysis focuses on the expression of RBM5 and compares it with the CFLAR expression. Moreover, RBM5 expression is also examined across tumor and normal tissue samples. This project presents summary statistics and several visualizations to characterize the data and identify patterns.

2 Methods

The analysis was conducted using R and with the use of RNA-sequencing count data and associated sample metadata, which were imported from CSV files. The gene of focus was

RBM5, while CFLAR was selected as the second gene for comparison. The chosen covariate was tissue type, and was categorized as tumor or normal.

Data visualizations were created using the following R packages: tidyverse (version 2.0.0), ggplot2 (version 4.0.3), ComplexHeatmap (version 2.28.0), circlize (version0.4.18), gridExtra (version 2.3.1), and ggpubr (version 1.0.0). The first visualization displayed is a histogram, which shows the distribution of RBM5 expression. Next, a scatter plot was created to exhibit the relationship between RBM5 and CFLAR expression. A violin plot and cumulative distribution function were also generated to compare RBM5 expression across normal and tumor samples. In addition, a box plot was created to compare the median and variability of RBM5 expression between the two different tissue types. Finally, the ComplexHeatmap package was used to create a heatmap to display expression counts for 10 selected genes.

3 Results

3.1 Summary Statistics

The summary statistics showcase the differences in the expression counts between RBM5 and CFLAR across the samples. CFLAR has a higher mean expression count (5,442), while RBM5 has a mean expression count of 4,044. Additionally, CFLAR has a higher median of 4,799 compared to RBM5's of 3,644. There is a higher standard deviation with CFLAR than RBM5, which indicates that CFLAR had higher expression counts and greater variation across the samples.

3.2 Histogram

The histogram reveals the distribution of RBM5 expression counts across the samples. The plot displays how most samples have expression counts between 2,000 and 5,000. The highest frequency occurs at around 3,000 expression counts. The distribution is right skewed with fewer samples having higher expression counts.

3.3 Scatter plot

The scatter plot shows the relationship between RBM5 and CFLAR expression counts. The plot illustrates an upward trend, which suggests a positive association between the expression counts of RBM5 and CFLAR. Samples with higher RBM5 counts often also have higher CFLAR counts. There is also considerable variation observed.

	RBM5	CFLAR
Mean	4,044	5,442
Median	3,664	4,799
1st Quartile	2,658	3,346
3rd Quartile	5,116	6,720
Min	671	664
Max	12,672	21,182
Standard Deviation	1,901.176	3,040.001

Figure 1: Summary Statistics of RBM5 and CFLAR

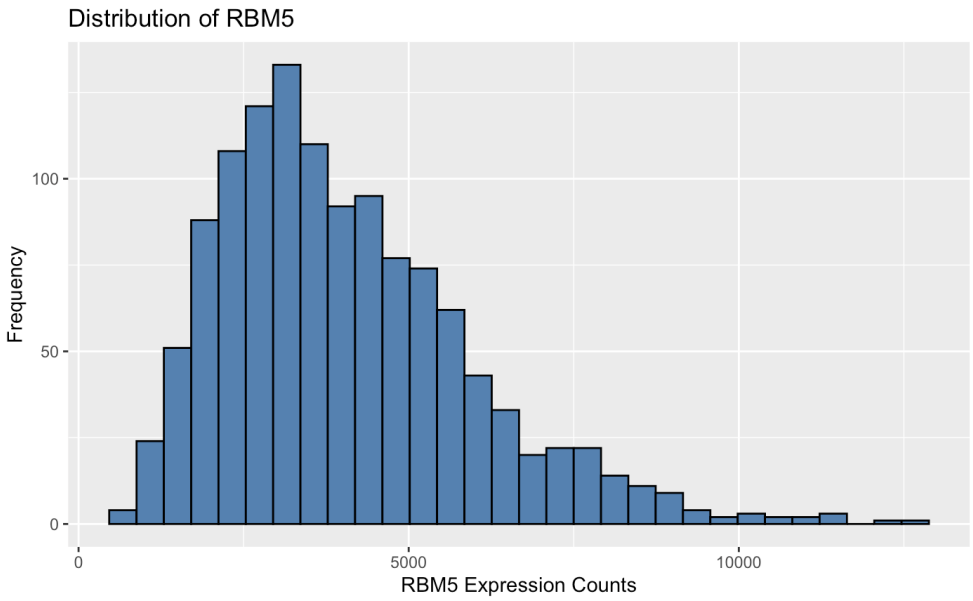


Figure 2: Distribution of RBM5 Expression Counts.

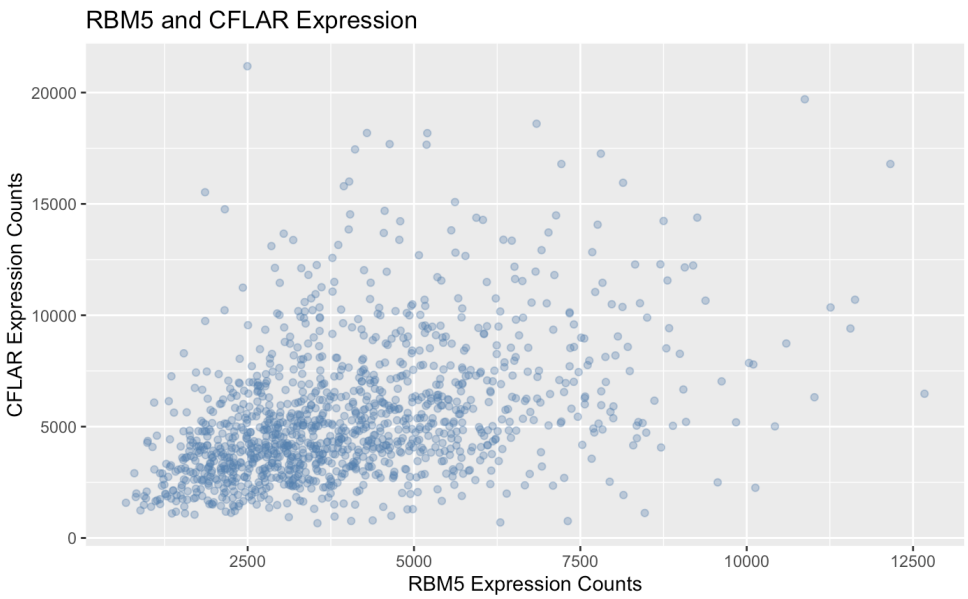


Figure 3: Scatter plot of the relationship between RBM5 and CFLAR expression counts.

3.4 Violin Plot

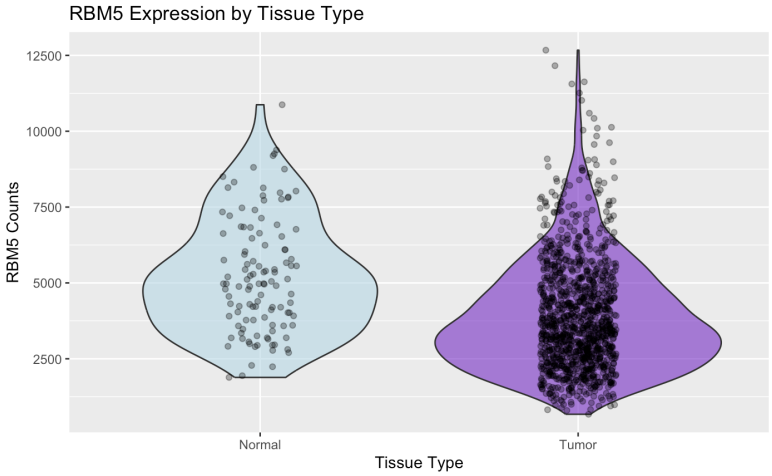


Figure 4: Distribution of RBM5 expression counts in normal and tumor tissues.

The violin plot compares RBM5 expression counts between normal and tumor samples. The tumor group contains more plotted observations than the normal group. The plot shows a greater concentration of counts around 3,000-5,000 for normal tissue samples. For tumor samples, there is a larger concentration of counts around 2,000-4,000. Additionally, there is a longer upper tail that extends beyond 10,000 for tumor samples. The plot showcases there are more observations at lower expression counts for tumor tissues.

3.5 Box Plot

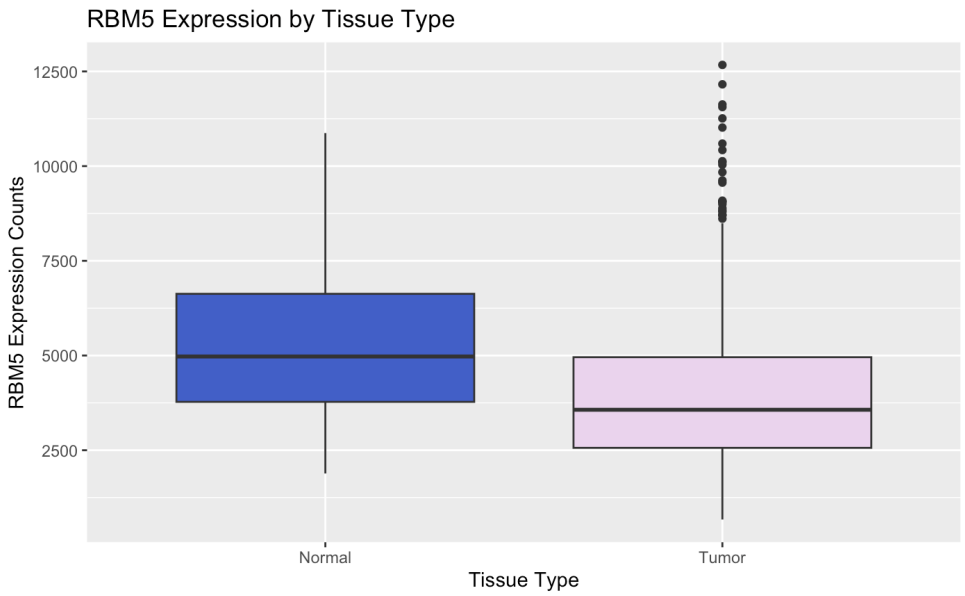


Figure 5: Comparison of RBM5 expression counts between normal and tumor tissue samples.

The boxplot shown compares RBM5 expression counts between normal and tumor tissue samples. Normal samples exhibit a higher median count of about 5,000 compared with 3,600 for tumor samples. For the tumor group, there are several high expression outliers. Furthermore, the middle 50 percent of normal samples show a higher expression range than tumor samples. These observations indicate RBM5 counts tend to be lower in tumor samples.

3.6 Heatmap

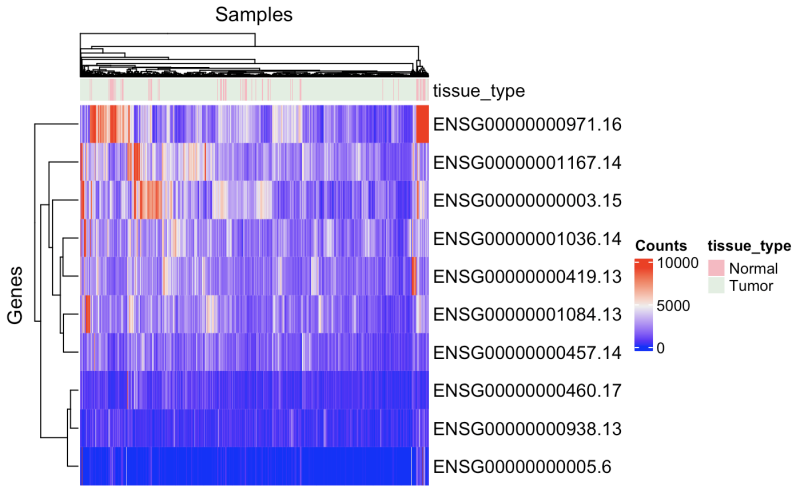


Figure 6: Expression counts of ten genes.

The heatmap displays the expression counts for 10 genes across the samples. On the color scale, shades of blue represent lower counts, while shades of red represent higher counts. Visually, some of the genes appear to have more instances of higher expression counts among normal tissue samples; however, this does not remain consistent for all genes. The tissue type annotation identifies normal and tumor samples to allow for comparison with the counts of the other genes.

3.7 CDF

The cumulative distribution function compares RBM5 expression counts between normal and tumor tissue samples. The plot shows how the tumor curve rises quicker, which indicates that tumor samples have lower RBM5 counts. Overall, the CDF reveals evidence of difference in RBM5 count distributions between normal and tumor tissue types.

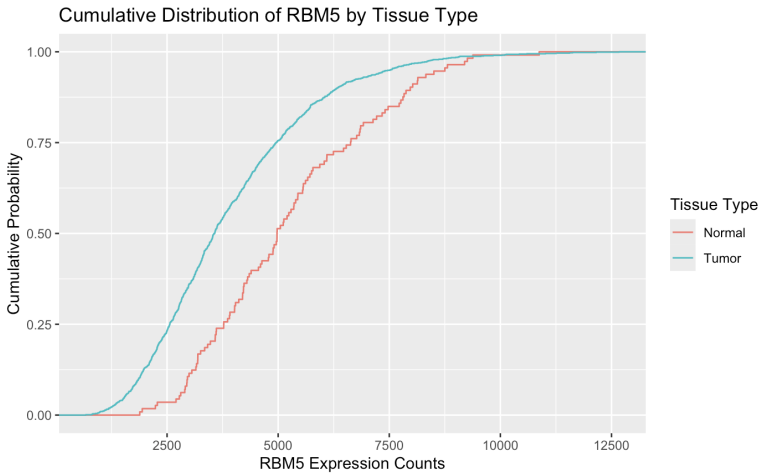


Figure 7: Cumulative distribution of RBM5 expression counts.

References

[1] National Center for Biotechnology Information. (2026, August 5). *RBM5 RNA binding motif protein 5 [Homo sapiens (human)].* NCBI Gene. `https://www.ncbi.nlm.nih.gov/gene/10181`

[2] Zhang, H., Li, X., & Lin, L. (2025). Biological functions and clinical implications of CFLAR: From cell death mechanisms to therapeutic targeting in immune regulation. *Journal of Inflammation Research, 18*, 4911–4928. `https://doi.org/10.2147/JIR.S519885`

[3] National Cancer Insitute (NCI). (n.d.). *The Cancer Genome Atlas Program*